

NyayaSetu Legal Benchmark (NSLB) v1.0

Benchmark Characterization Report

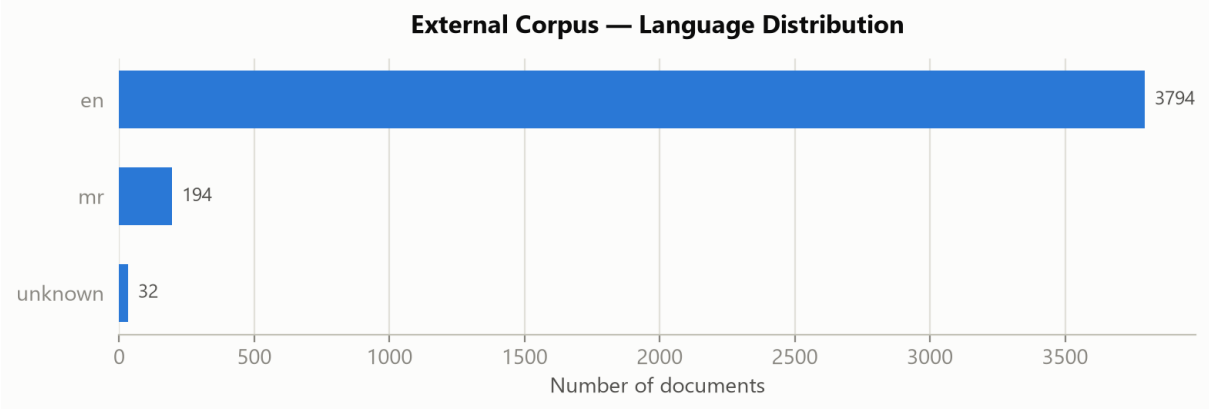
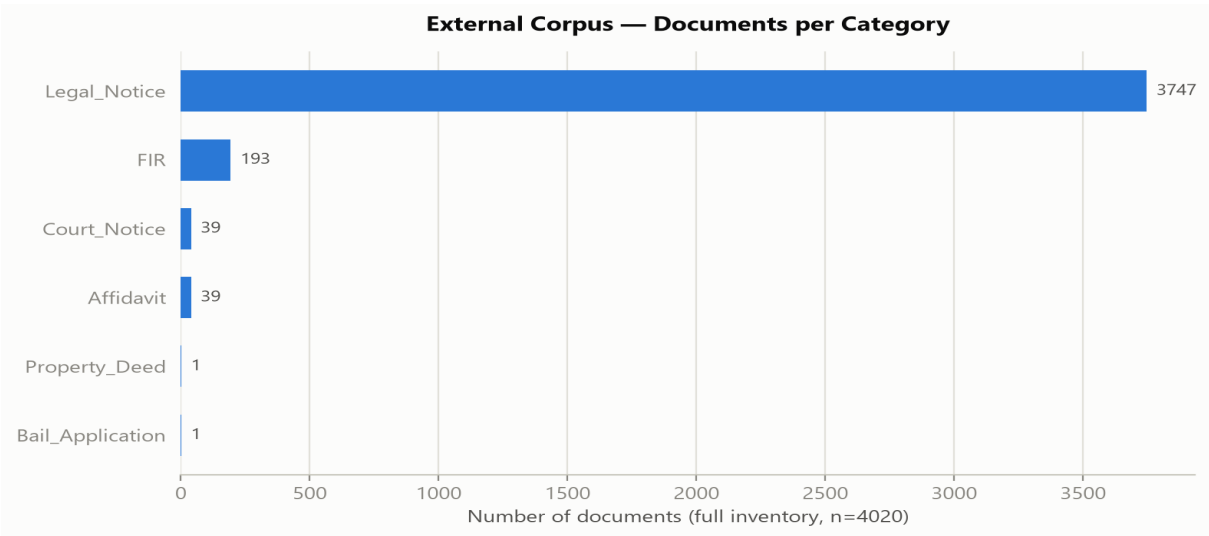
477 records across 9 datasets, built from 1 phase-1 in-repo source plus 94 deep-processed documents drawn from a 4020-document external corpus (fully inventoried). No evaluation experiments have been run — this document characterizes the ground-truth benchmark itself.

1. Overall Benchmark Summary

Metric	Value
Total source documents inventoried (external corpus)	4020
Total source documents (phase-1 in-repo fixtures)	1
Total documents deep-processed (external corpus)	94
Total documents deep-processed (all sources)	95
Total benchmark records	477
Number of datasets	9
Number of document categories (external corpus)	6
Number of document-type labels observed in document_analysis	6
Number of supported languages (system-wide)	13
Number of languages with real content in NSLB	2
Number of distinct IPC sections (verified mapping)	171
Number of distinct CrPC sections (verified mapping)	12
Number of distinct IEA sections (verified mapping)	2
Number of distinct BNS target sections	148
Number of distinct BNSS target sections	12
Number of distinct BSA target sections	2
Number of abolished (decriminalized) sections	2
Number of retrieval queries	46
Number of QA pairs	67
Number of document-analysis records	90
Number of IPC->BNS mapping records	185
Number of multilingual records	12
Number of robustness records	11
Number of end-to-end evaluation cases	11
Number of research queries	43
Number of translation records	12

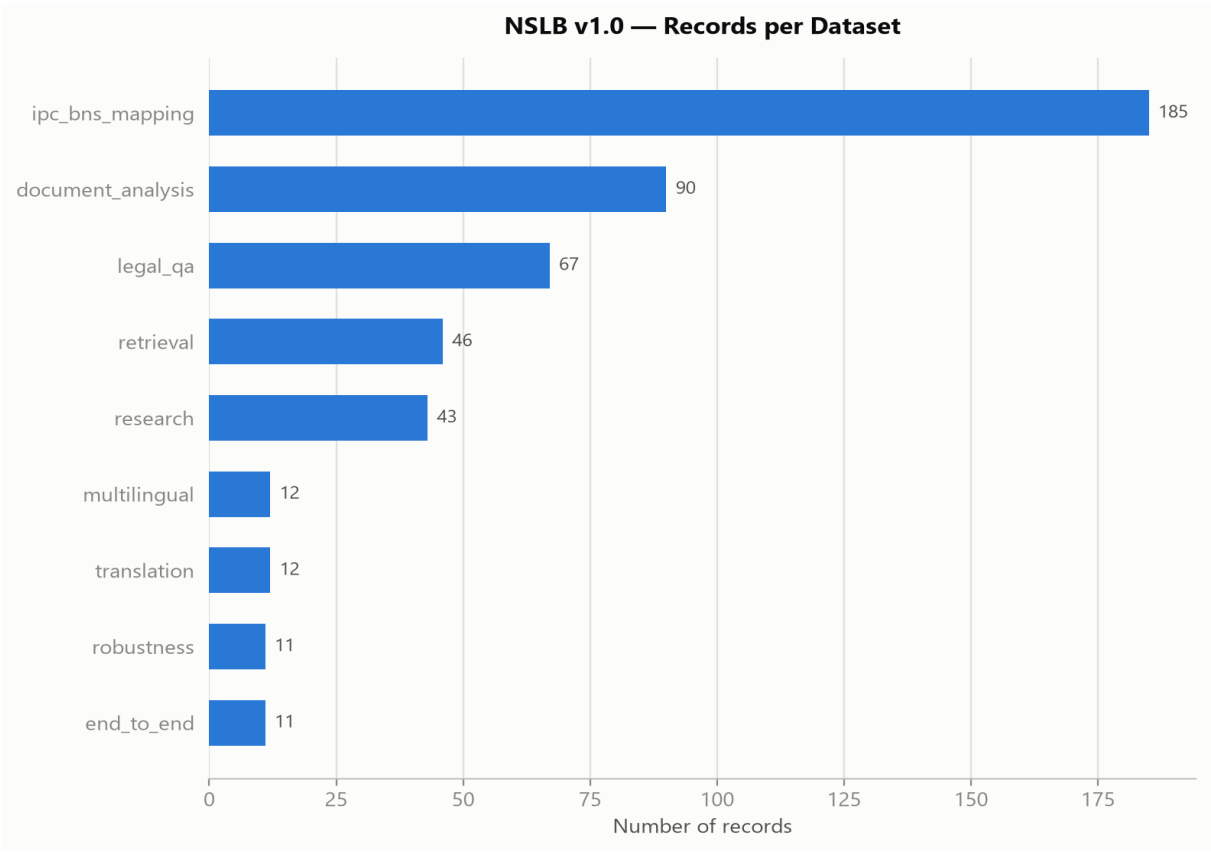
2. Corpus Statistics

Category	Documents	Avg Pages	Median Pages	Min Pages	Max Pages	OCR-Required	Empty	Unreadable	Top Language
Affidavit	39	23.87	21	13	70	31	31	0	unknown
Bail_Application	1	66	66	66	66	0	0	0	en
Court_Notice	39	27.95	26	7	76	0	0	0	en
FIR	193	6.88	6	6	20	0	0	0	mr
Legal_Notice	3747	6.33	3	1	104	0	0	0	en
Property_Deed	1	25	25	25	25	1	1	0	unknown
TOTAL	4020	6.75	3.0	1	104	32	32	0	-

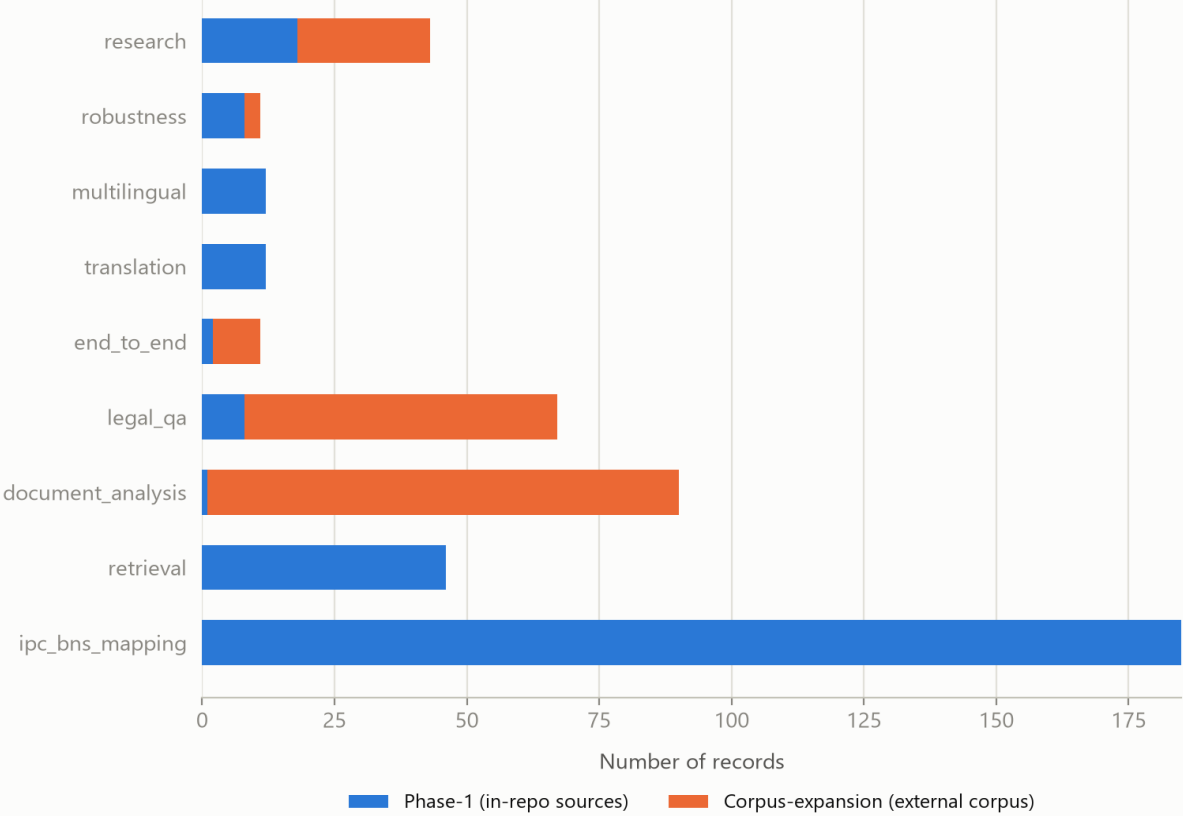


3. Dataset Statistics

Dataset	Records	% of Benchmark	Phase-1	Corpus-Expansion	Avg. Fields Populated	Completeness	Validation Status
ipc_bns_mapping	185	38.78%	185	0	7.0	87.5%	PASS
retrieval	46	9.64%	46	0	5.0	83.3%	PASS
document_analysis	90	18.87%	1	89	5.08	72.5%	PASS
legal_qa	67	14.05%	8	59	7.85	98.1%	PASS
end_to_end	11	2.31%	2	9	6.82	97.4%	PASS
translation	12	2.52%	12	0	6.0	75.0%	PASS
multilingual	12	2.52%	12	0	7.0	77.8%	PASS
robustness	11	2.31%	8	3	6.27	89.6%	PASS
research	43	9.01%	18	25	5.02	83.7%	PASS
TOTAL	477	100%	292	185	-	-	-

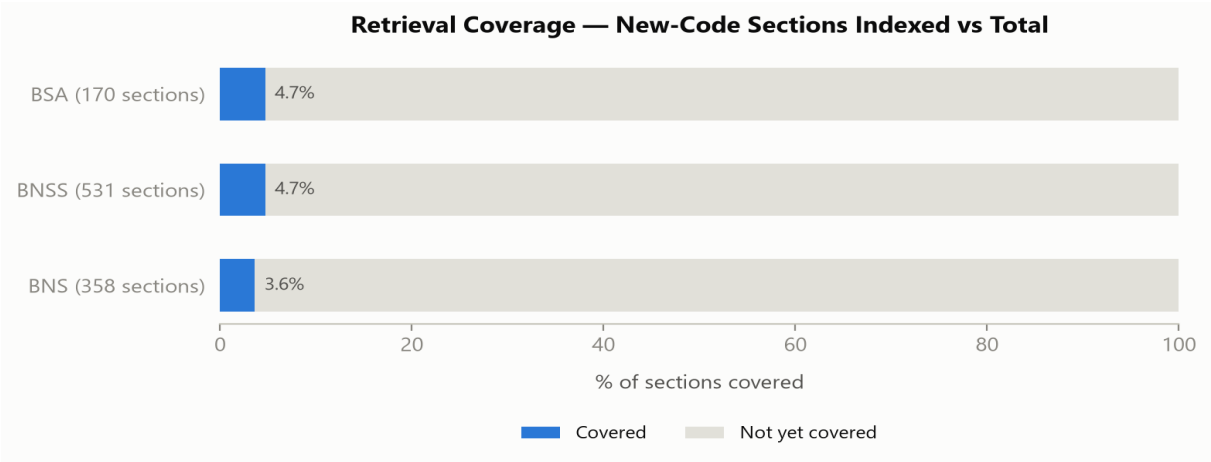


NSLB v1.0 — Dataset Composition by Source Phase



4. Coverage Statistics

Coverage Dimension	Covered	Total (in-benchmark denominator)	Coverage %
BNS sections (retrieval-indexed)	13	358	3.63%
BNSS sections (retrieval-indexed)	25	531	4.71%
BSA sections (retrieval-indexed)	8	170	4.71%
Combined new-code sections (BNS+BNSS+BSA)	46	1059	4.34%
ChromaDB chunk coverage (retrieval.jsonl)	46	3254	1.41%
Corpus citation instances matched to known mapping	3	9	33.3%
Document coverage (deep-processed / inventoried)	94	4020	2.34%
Language coverage (real content / supported)	2	13	15.4%



Denominators for BNS/BNSS/BSA coverage (358/531/170) are each act's own final section number, already extracted verbatim from its repeal clause and stored in retrieval.jsonl (ids retr_bns_358, retr_bnss_531, retr_bsa_170) — not an imported fact. No authoritative IPC/CrPC/IEA total-section count exists in this project, so only absolute mapped-section counts are reported for those acts.

5. Data Quality Analysis

Quality Metric	Count
Duplicate records removed (generation-time)	5
Duplicate content records removed (validation-time)	9
Invalid records rejected by schema loader	0
Records with missing metadata (tags/notes)	0
Records with missing citations	0
Contradictory IPC->BNS mappings	0
Duplicate ids (current)	0
Duplicate QA content (current)	0
Total records checked	477
Validation success rate	100.0%

Validation success rate: 100.0% across 477 records, independently re-checked by this report (duplicate ids, duplicate QA content, contradictory mappings, missing metadata) rather than only citing the build-time log.

6. Benchmark Characteristics

Strengths

Every record traces to a real, named source; 100.0% validation success rate; broad absolute IPC/CrPC/IEA mapping coverage merged from two independent sources.

Limitations

Zero contract-type document coverage (rental/employment/loan/sale/service/NDA); 2/13 language coverage; 4.34% retrieval section coverage; legal_qa corpus-expansion records are entirely template-generated, not human-authored.

Potential bias / imbalance

Corpus is 93%+ Legal_Notice documents concentrated in 2 of 3 High Court sources; document_analysis is almost entirely FIR/Legal_Notice/Court_Notice/Affidavit; language content is English/Marathi only.

Remaining work for publication

~20-25 labeled contract-type documents; ~30-40 human-authored comprehension QA records; a separate multilingual source-acquisition effort; a section-aware statute re-chunker (or ~50-100 manually labeled retrieval queries) to grow retrieval coverage.

Full detail, all 5 tables as CSV, and all 7 figures in PNG+SVG: see [evaluation/reports/benchmark_summary.md](#), [evaluation/reports/tables/](#), and [evaluation/reports/figures/](#).